Biochemistry Cume
December 4, 2009

1. An amino acid mixture consisting of lysine, leucine, and glutamate is to be separated by ion-exchange chromatography, using a cation-exchange resin at pH 3.5. The eluting buffer is also pH 3.5. Which of these amino acids will be eluted from the column first? Will any other treatment be needed to elute one of these amino acids from the column? Justify your answers. (25 points)

2. An amino acid mixture consisting of D, G, and F is to be separated by HPLC. The stationary phase is hydrophilic and the mobile phase is a solvent less polar than water. Which of these amino acids will move the fastest? Which one will move the slowest? Justify your answer. (25 points)

3. Gel-filtration chromatography is a useful method for removing salts, such as ammonium sulfate, from protein solutions. Describe how such a separation is accomplished?(25 points)

4. Describe materials that would be necessary and sufficient to purify a protein by means of affinity chromatography. (25 points)